06 · Is spend *causing* sales? — causal MMM (pymc-marketing)

July 12, 2026

> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The business decision. Brand search converts like crazy — should we pour budget into it? Be careful: brand search mostly **catches** demand that upper-funnel channels (TV) and seasonality already created, so crediting it for those conversions and shifting budget its way would be a mistake.

0.0.1 The concepts

- **MMM (Marketing Mix Model)** — a regression of sales on each channel's spend over time (usually weekly), used to attribute sales to channels and set budgets. It's the "top-down" alternative to click-tracking.
- **Adstock (carry-over)** — advertising doesn't only work the week it runs; its effect *decays over subsequent weeks*. Adstock transforms raw spend into an "effective" exposure that carries over.
- **Saturation (diminishing returns)** — the 10th €1,000 into a channel buys less lift than the 1st. A saturation curve bends over as spend rises; its *slope* is the **marginal** return that should drive the next euro of budget.
- **Last-click attribution** — crediting the conversion to the final touch (often brand search). It systematically **over-credits** demand-capture channels, which is the error a causal MMM aims to fix.
- **Incremental vs mediated** — the *incremental* contribution of a channel is what it *caused*; a channel can show a big correlation with sales while causing very little (it was downstream of the real driver).

0.0.2 The arc of this notebook — a cautionary tale with a resolution

A plain MMM regresses sales on all channels and is fooled by a **confounder** like seasonality (it lifts sales *and* drives brand-search spend). The textbook fix is a **causal MMM**: hand it a **DAG** and let the backdoor logic (nb 05) control for the confounder. This notebook shows that fix is **necessary but not sufficient**, in three acts:

1. **The confounding correction works — in a linear model.** A naive OLS attribution that omits seasonality massively over-credits brand_search; adding seasonality as a control collapses that credit toward the small truth. That's the cleanest way to *see* the confounding, and it's robust.
2. **The out-of-the-box causal MMM still inverts the ranking.** Fit with library-default priors, the MMM's *flexible saturation curve* for brand_search **re-absorbs** the seasonal demand the linear control was meant to remove — so it credits brand_search *more* than TV,

the opposite of the truth. A DAG plus a linear control does **not** de-confound a channel whose spend is collinear with the confounder.

3. **Experiment-calibrated priors recover it.** Feed in what a small **geo experiment** teaches — that brand_search's incremental effect is small — as an informative prior on its saturation, and the MMM ranks the channels correctly again. **This is why MMMs must be calibrated with experiments** (Anchor B, nb 07): observational data alone cannot pin the decomposition.

   On real data. Swap in your **own weekly spend-by-channel + sales** table plus known confounders (seasonality, price, distribution). Public example: the datasets shipped with Meta's *Robyn* or `pymc-marketing`'s own tutorials. Always pair an MMM with occasional geo experiments to anchor the levels.

It follows the repo's fixed **7-step contract**: question -> simulate -> identify -> estimate -> validate -> decide in € -> caveats.

0.1 2 · Simulate a ground truth

Weekly sales from a faithful **adstock + Michaelis-Menten saturation** process. **TV** is a big genuine driver. **seasonality** is a **confounder** — it lifts sales directly *and* drives brand-search spend. **brand_search** has only a *small* real effect, but its spend rides the seasonal wave (corr ~ 0.75), so a naive attribution ignoring seasonality massively over-credits it. Honest answer: TV highly incremental, brand_search only slightly.

The data-generating model — exactly what `dgp.mmm_weekly` implements (defaults & seed in `src/cmp/dgp.py`). Weeks $t = 0, \dots, 155$; seasonality $s_t = \sin(2\pi t/52)$:

$$x_t^{\text{tv}} = \max\{1,\ 40 + 8\, s_t + \varepsilon_t^{\text{tv}}\}, \quad \varepsilon_t^{\text{tv}} \sim \mathcal{N}(0, 8^2); \qquad x_t^{\text{bs}} = \max\{1,\ 25 + 11\, s_t + \varepsilon_t^{\text{bs}}\}, \quad \varepsilon_t^{\text{bs}} \sim \mathcal{N}(0, 7^2)$$

$$a_t(x;\lambda) = x_t + \lambda\, a_{t-1}(x;\lambda) \qquad \text{(geometric adstock / carry-over)}$$

$$f(a;\alpha,\kappa) = \frac{\alpha\, a}{\kappa + a} \qquad \text{(Michaelis–Menten saturation)}$$

$$\text{sales}_t = 100\ +\ \underbrace{f(a_t(x^{\text{tv}};\,0.4);\ 60,\ 50)}_{\text{TV: big, real}}\ +\ \underbrace{f(a_t(x^{\text{bs}};\,0.2);\ 8,\ 40)}_{\text{brand search: small, real}}\ +\ \underbrace{35\, s_t}_{\text{season}\to\text{sales}}\ + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,\ 4^2).$$

The confounding is the **two appearances of** s_t: it drives brand-search spend (the $11\, s_t$ term — people search the brand more in peak season; corr ~ 0.75) *and* sales directly (the $35\, s_t$ term). An MMM that omits seasonality hands much of that $35\, s_t$ to brand search. Note the generator uses the MMM's own functional form (adstock -> saturation), so the model is well-specified — any remaining failure is about *identification*, not fit. **Identification** is whether the data can, even in principle, pin down the answer uniquely: a model can fit the sales series perfectly and still leave the *split* of credit between two channels undetermined — that gap between 'fits' and 'is identified' is the whole subject of this notebook.

```
TRUE incremental sales - TV 3,536   brand_search 340 (small)

    date_week         tv  brand_search  seasonality       sales
0  2023-01-02  42.870187     17.416271     0.000000  130.484054
1  2023-01-09  53.049712      5.742880     0.120537  135.016128
```

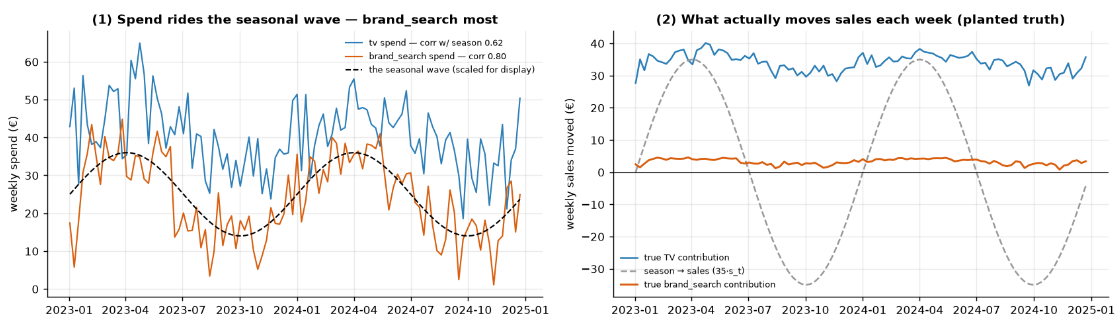
```
2 2023-01-16  27.623875     17.944274     0.239316  134.429980
3 2023-01-23  56.329747     30.742142     0.354605  152.430146
4 2023-01-30  43.339248     35.779100     0.464723  156.900956
```

0.1.1 First — look at the data: the confounding, made visible

Two pictures to hold in your head for the rest of the notebook, drawn **before any model sees the data**:

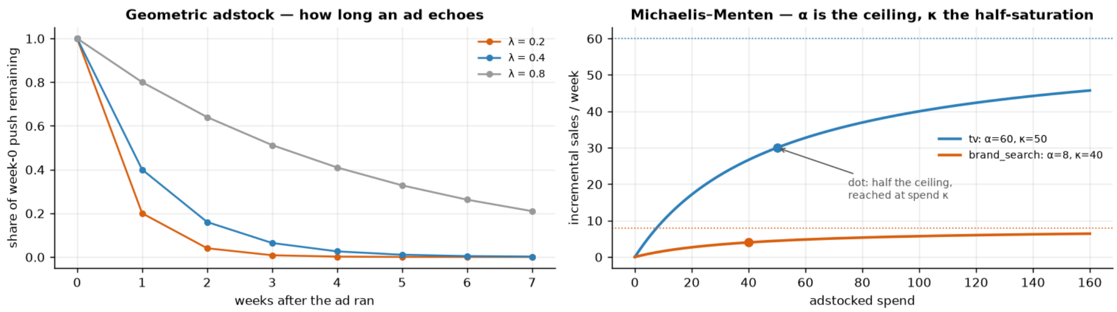
1. **Spend rides the seasonal wave.** brand_search's spend is *built* from the same seasonal index that lifts sales, so it tracks the dashed wave far more tightly than TV does (both correlations are printed in the legend). Whatever the season does to sales will look, to a model that ignores the season, like brand_search doing it.
2. **Most of what brand_search appears to move is the season.** The right panel plots the *planted* weekly components of sales: TV's contribution is large, the +/-35 seasonal swing is larger still — and brand_search's true contribution is the near-flat line hugging the bottom. An attribution method must hand the wave back to the season; everything that follows is about whether it does.

```
corr(spend, seasonality): brand_search 0.80 vs tv 0.62 - the confounder is
entangled
with brand_search; true weekly contributions: TV ~ €34, brand_search ~ €3,
season swing +/-€35.
```



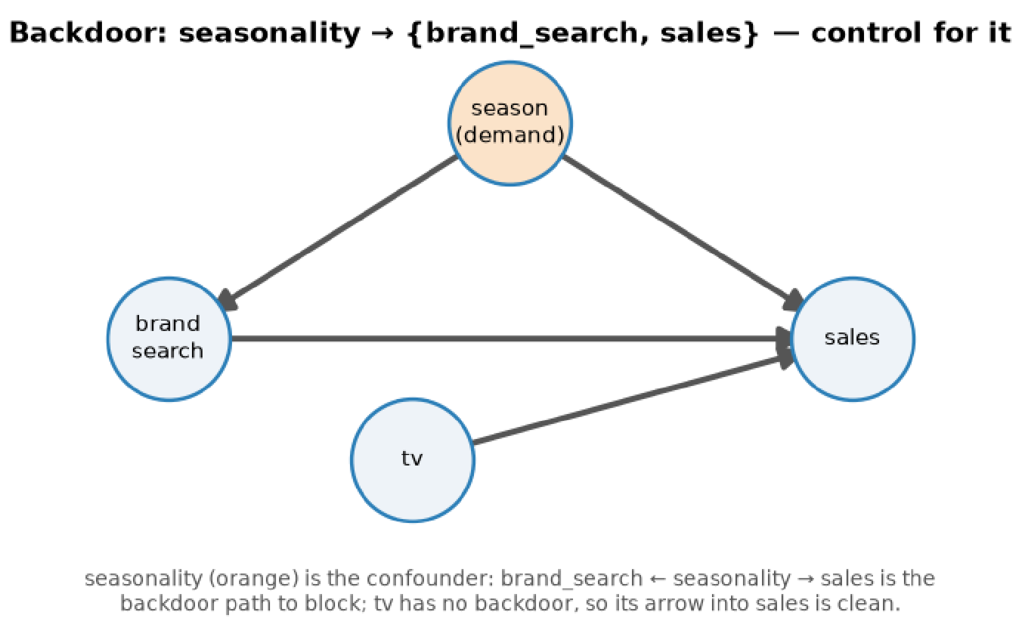
0.1.2 The two transforms, before we use them in anger

The MMM's channel response is built from the two primitives the generator above already used — worth ten seconds of intuition each before a sampler spends minutes on them. **Adstock** answers *"how long does an ad echo?"*: with geometric decay λ, a euro of spend still exerts λ^{ℓ} of its initial push ℓ weeks later, so $\lambda = 0.2$ is a one-week echo while $\lambda = 0.8$ still echoes seven weeks out. **Saturation** answers *"what is the next euro worth?"*: the Michaelis–Menten curve $f(a) = \alpha a/(\kappa + a)$ climbs toward a **ceiling** α — the most the channel could ever add per week — and passes half of it exactly at $a = \kappa$, the **half-saturation** point. Keep the right panel in mind for Step 4b: the experiment-calibrated prior acts on the *ceiling* α, the parameter that says how much a channel could possibly deliver — precisely the number a lift test measures.



0.2 3 · Identify — the marketing DAG picks what to control for

DAG: `seasonality -> {sales, brand_search}`, `tv -> sales`, `brand_search -> sales`. The backdoor criterion says **seasonality is a confounder -> control for it**; omitting it is exactly the last-click mistake that over-credits brand search. (This DAG has a confounder but no **mediator** — a variable sitting *on* the causal path from spend to sales, which you would deliberately *not* control for; nb 05 covers that distinction.) The **estimand** — the precise quantity we want to estimate — for channel c is its **incremental contribution** — sum over weeks of $\mathbb{E}[\text{sales} \mid do(\text{spend}_c = \text{observed})] - \mathbb{E}[\text{sales} \mid do(\text{spend}_c = 0)]$, holding the other channels at their observed spend (how much of sales would vanish if we switched channel c off). Here $do(\cdot)$ denotes an **intervention** — setting spend by decree rather than merely observing it (nb 05). **ROAS** (return on ad spend) = incremental contribution ÷ spend, each with a posterior — note this is *revenue*, not profit, so at 1x you break even only on a 100% margin (we return to margins in Step 6). Identification comes from the backdoor set {seasonality}; we hand the DAG to the MMM via `dag`, `treatment_nodes`, `outcome_node`.



```
Backdoor adjustment set the causal layer will use: ['seasonality']
```

0.3 4 · Estimate — fit the causal MMM twice

We fit the **same DAG-aware MMM twice**, to make the lesson unmissable:

- **(a) default priors** — the honest, out-of-the-box causal MMM. Watch it *invert* the ranking.
- **(b) experiment-calibrated priors** — encode what a geo experiment teaches (brand_search's incremental effect is small) as an informative prior on its saturation ceiling, and watch the ranking snap back.

The fitted model, in symbols. Per channel $c \in \{\text{tv}, \text{brand\_search}\}$, spend is adstocked over an 8-week window, saturated, and summed with the control:

$$\text{sales}_t \sim \mathcal{N}\Big(\beta_0 + \textstyle\sum_c f_c(a_c(x^c)_t) + \gamma\, s_t,\ \ \sigma^2\Big), \qquad a_c(x)_t \propto \textstyle\sum_{\ell=0}^{7} \lambda_c^{\ell}\, x_{t-\ell}, \qquad f_c(a) = \tfrac{\alpha_c\, a}{\kappa_c + a},$$

geometric adstock (`GeometricAdstock(l_max=8)`) and Michaelis–Menten saturation, with each channel's decay λ_c, ceiling α_c and half-saturation κ_c **estimated** from the data. Fit (a) leaves their priors at library defaults; fit (b) replaces the prior on brand-search's ceiling α_{bs} with the experiment-informed one. Note the functional form matches step 2's generator exactly — this is the *well-specified best case*, which is what makes the ranking inversion in (a) a lesson about identification rather than model misfit. (pymc-marketing scales variables internally; the equations are written in original units for readability.)

The priors, made explicit. "Library defaults" in fit (a) means, per channel c (all on **max-abs-scaled** spend and sales — pymc-marketing divides each channel's spend and the sales series by its own maximum before fitting; Step 4b returns to why that matters):

$$\lambda_c \sim \text{Beta}(1,\,3), \qquad \alpha_c \sim \text{Gamma}(\mu = 2,\ \sigma = 1), \qquad \kappa_c \sim \text{HalfNormal}(1),$$
$$\beta_0,\, \gamma \sim \mathcal{N}(0,\,2), \qquad \sigma_y \sim \text{HalfNormal}(2).$$

The one to stare at is the saturation **ceiling** α_c, in scaled-sales units: a prior mean of 2 says "this channel could plausibly add up to about *twice the all-time weekly sales peak*, every week" — weakly-informative in the worst sense, generous exactly along the direction the data cannot push back on. Fit (b) replaces only brand-search's copy of that one prior.

Convergence housekeeping: this is the hardest posterior geometry in the cookbook — the FULL profile needs `target_accept = 0.99` and deeper trees to reach R-hat ~ 1.01 (the diagnostics — R-hat, ESS, divergences — are defined in notebook 01's "did the sampler converge?" box). Even then, expect fit (a) to log a **handful of divergences** and fit (b) to sit exactly at the 1.01 edge — and in FAST mode either fit's short chains can disagree outright (R-hat » 1.01). Read those complaints as **information, not nuisance**: the sampler is struggling along the very season/brand-search ridge — a flat valley of near-equivalent answers — that breaks identification (we show it in the flesh right after the fit). We quote fit (a) only for its — robustly wrong — ranking, never for its intervals.

A caution that runs through both: **MMMs are miscalibrated in *absolute* terms** — with channel spend riding seasonal demand, a short weekly series can't fully separate "channel drove sales" from "season drove both", so fitted contributions run high. What we're after is the **ranking** and each channel's **ROI verdict**.

```
(a) default-prior MMM convergence: max r-hat 2.150 - min ESS 3 - divergences 207
(a) default contribution - TV 15,547   brand_search 22,962
    true: TV 3,536, brand_search 340  ->  the MMM ranks brand_search ABOVE TV
(inverted!).
```

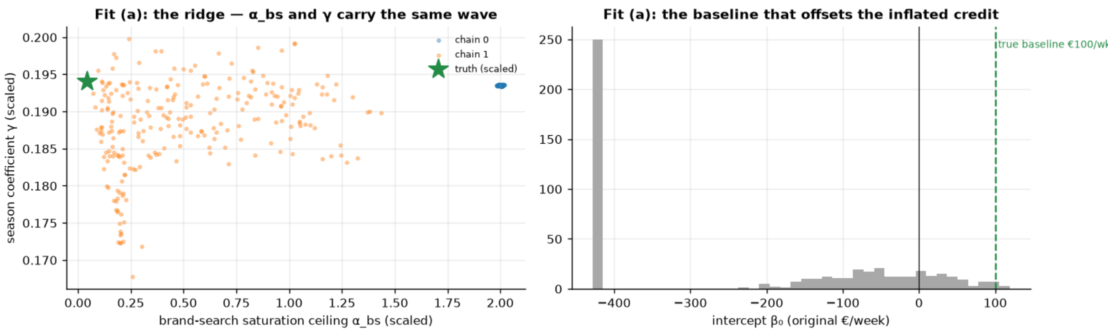
0.3.1 Why the out-of-the-box MMM inverts the ranking

The default MMM credits brand_search *above* TV — backwards. The mechanism is worth seeing, because it's a real trap:

- brand_search's spend is *built* to ride the seasonal demand wave (Step 2 constructs it from the same seasonal index that drives sales), and the fit lets its flexible saturation curve quietly **re-absorb the seasonal demand** the linear `seasonality` control was meant to remove.
- Because brand_search rides the seasonal wave more tightly than TV does, it wins that tug-of-war and books a huge contribution (tens of thousands vs a true ~500), while a large **negative baseline** offsets it — the classic broken **level decomposition** — how the model splits the *absolute* level of sales across baseline + channels — of a non-identified MMM.
- The DAG was right and the control was present; they still weren't enough. **A linear confounder-control does not de-confound a channel whose spend is** collinear** with (moves almost in lockstep with) the confounder and enters through a flexible functional form.** Observational data alone can't break the tie.

Those are three checkable claims — collinear spend, a posterior tug-of-war, a negative baseline — so the next cell checks them, in the data and in fit (a)'s own posterior.

```
spend-confounder correlations (the tv row is the contrast):
                tv  brand_search  seasonality
tv            1.00          0.48         0.62
brand_search  0.48          1.00         0.80
seasonality   0.62          0.80         1.00

draw-level corr(alpha_bs, gamma) = +0.58   ·   per-chain alpha_bs posterior
means: [2.   0.52]
truth in scaled units: alpha_bs ~ 0.044, gamma ~ 0.194 - the green star, far
from the mass.
intercept (original scale): mean €-234/week vs planted baseline €100/week  ->
NEGATIVE: the level decomposition is broken.
Note the per-chain means: the chains sit on DIFFERENT stretches of the ridge -
each found its own
split of the same wave and never reconciled. That is the R-hat warning above,
drawn as a picture.
```



Read-out. *Left*: every dot is a posterior draw of (brand-search ceiling α_{bs}, season coefficient γ). The truth (green star) has a tiny ceiling; the posterior instead spreads along a **ridge** of near-equivalent decompositions — draws (and, at FAST settings, whole chains) trade the seasonal wave back and forth between the two parameters, because the likelihood is near-flat along that direction. *Right*: to pay for the inflated brand-search credit, the intercept is dragged far below the planted €100/week — the classic broken level decomposition of a non-identified MMM, exactly the "large negative baseline" claimed above. When two regressors carry the same wave, the posterior does not pick one: it spreads over every split the data cannot distinguish, and under the default priors the flexible saturation curve books the credit. **No amount of extra sampling fixes this — only information from outside the dataset can.** Which is the move of the next section.

0.3.2 The fix — priors calibrated from a geo experiment

The standard remedy is exactly the discipline of Anchor B (the cookbook's 'calibrate MMMs with experiments' rule, nb 07): run a **small geo experiment** — which randomizes a channel's spend across matched groups of regions (some markets get the ads, some don't) and reads the sales gap as that channel's *true* incremental lift, uncontaminated by the seasonality the observational data can't net out (full treatment in nb 07) — to learn that brand_search's *incremental* effect is small, and feed that back as an **informative prior on its saturation ceiling** (`saturation_alpha`). TV, whose incremental role we haven't independently pinned down, keeps a weakly-informative prior. This is not putting a thumb on the scale — it's using experimental evidence exactly where observational data is silent.

From experimental euros to a prior — the mapping, explicitly. The geo test (nb 07) delivers an estimate of brand-search's total incremental sales over its test window, $\hat{L} \pm \mathrm{SE}$. The MMM says brand-search's weekly contribution at the observed operating point is $f(\bar{a}) = \alpha_{bs}\,\bar{a}/(\kappa_{bs} + \bar{a})$, with $\bar{a}$ the typical adstocked spend. Matching the model to the experiment therefore pins the ceiling:

$$\alpha_{bs} \;\approx\; \underbrace{\frac{\hat{L}}{n_{\text{weeks}}}}_{\text{weekly lift the test measured}} \cdot \underbrace{\frac{\kappa_{bs} + \bar{a}}{\bar{a}}}_{\text{ceiling from the operating point}}, \qquad \alpha_{bs}^{(s)} \sim \mathrm{Gamma}(\mu = 0.3,\, \sigma = 0.15),$$

with σ scaled from the experiment's SE — wide enough that the MMM's data may still disagree, tight enough to break the ridge. One unit convention to keep straight: pymc-marketing fits on

max-abs-scaled spend and sales, so the prior acts on $\alpha_{bs}^{(s)} = \alpha_{bs}/\max_t(\text{sales}_t)$ — which is why the 0.3 is not comparable to the DGP's $\alpha_{bs} = 8$ (sales units) without that conversion. A mean of 0.3 says "the test showed brand search can move at most a modest fraction of peak weekly sales"; TV keeps the default Gamma($\mu = 2, \sigma = 1$) — an implied ceiling near *twice* peak weekly sales, i.e. essentially unconstrained, honest about the fact that no experiment has pinned TV down. Step 5's sensitivity sweep then shows the conclusion survives anywhere in the experiment-plausible range — the answer to "didn't you just pick the prior you wanted?".

> **Production note.** pymc-marketing can also ingest the experiment *directly*: `MMM.add_lift_test_measurements(df_lift)` appends each lift measurement to the likelihood as a distance penalty on the saturation curve's rise over the test's spend delta — a cleaner, multi-test-ready alternative to hand-translating one experiment into one Gamma prior, and the recommended pattern outside the classroom.

```
(b) calibrated-prior MMM convergence: max r-hat 1.210 - min ESS 8 - divergences
61
(b) calibrated contribution - TV 11,678   brand_search 1,790
    true: TV 3,536, brand_search 340  ->  ranking RESTORED: TV above
brand_search.
```

0.4 5 · Validate — fit vs identification, the correction, recovery, and robustness

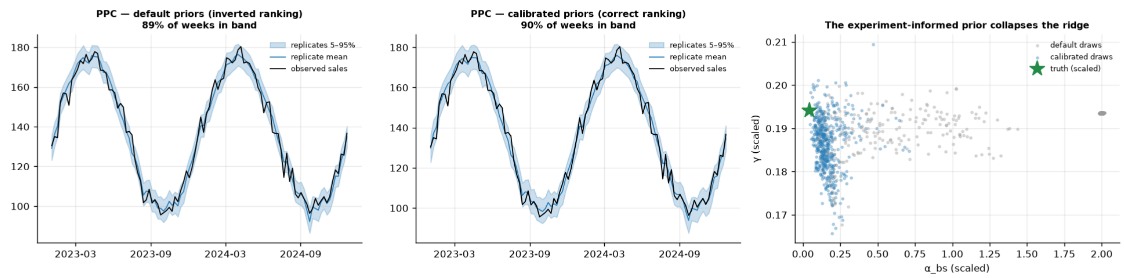
Six views, from "does it fit?" to "does the conclusion survive?":

1. **Posterior predictive check, both fits** — and the deliberately uncomfortable punchline that *fit quality cannot arbitrate between the inverted and the calibrated decomposition.*
2. **The confounding correction, several ways** — naive OLS (omit season) over-credits brand_search; OLS + **season** collapses it toward truth; the **default** causal MMM *inverts* it; the **calibrated** MMM restores the right ranking. The contrast between the two MMMs is the headline.
3. **ROAS with uncertainty (calibrated MMM)** — each channel's posterior ROAS vs the break-even line. ROAS = contribution ÷ (fixed **observed** spend), so it carries the **same relative miscalibration as the contribution** — dividing by a constant spend cannot remove the bias. What is robust is the ROAS **ranking** (and brand_search's below-break-even verdict), **not** the absolute ROAS *levels*, which are biased by the same factor as the contributions.
4. **Saturation curves** — contribution vs spend per channel, showing diminishing returns (the shape that sets *marginal* ROI and hence where the next euro should go).
5. **Parameter recovery vs the planted truth** — grade the *parameters*, not just the contributions, to see exactly where the non-identification lives (and why the calibrated prior is surgical).
6. **Prior sensitivity** — the fix as a robustness *range*, not a single hand-picked point.

0.4.1 View 1 — the posterior predictive check that cannot save you

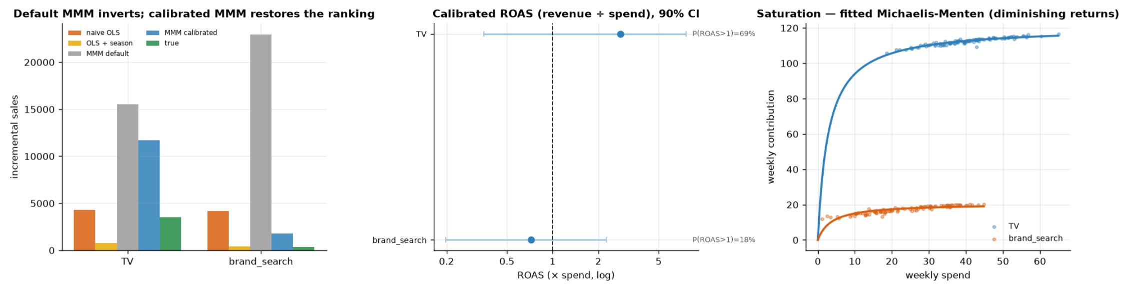
Before reading any decomposition off either fit, we ask the standard model-criticism question (nb 01, Step 4): *can the model reproduce the data it was trained on?* For each fit we draw replicated sales series from the posterior predictive, lay the replicates' 5–95% band over the observed series, and print the share of observed weeks inside the band (~90% means calibrated). Watch for what does **not** happen: the model with the inverted, wildly wrong decomposition does not fit any worse.

```
share of observed weeks inside the 5-95% replicate band - default 89% ·
calibrated 90%   (nominal ~ 90%)
replicate-mean RMSE (€/week)                              - default 3.5 ·
calibrated 3.4   (DGP noise sigma = 4)
Two decompositions that disagree by an order of magnitude on brand_search
reproduce the observed
sales series near-identically - the likelihood cannot tell them apart. The ridge
is where they hide.
```



Read-out. The two left panels are, for practical purposes, the same picture: near-identical bands, coverage and RMSE (printed above) — yet the left model credits brand_search with tens of times its true contribution. This is the precise sense in which the notebook's failure is **identification, not fit**: the generator uses the MMM's own functional form (Step 2's "well-specified" remark), so both fits sit near the same likelihood value, and no amount of posterior-predictive checking can distinguish the inverted decomposition from the correct one. Keep running PPCs — a model that *cannot* reproduce the series is disqualified before you read anything off it — but treat them as **necessary, never sufficient** for a causal claim. *Right panel*: what the experiment-informed prior actually did. The default posterior (grey) spreads along the ridge; the calibrated posterior (blue) is pinned at the low-ceiling end, beside the truth. The prior did not overrule the data — it **selected among answers the data could not rank**.

```
brand_search credit - naive 4,196 | OLS+season 399 | MMM default 22,962 | MMM
calibrated 1,790 | true 340
calibrated ROAS - TV 2.79x [90% 0.35,7.61]  ·  brand_search 0.72x [90%
0.20,2.27]
TRUE ROAS       - TV 0.84x · brand_search 0.14x  (both below 1x: the MMM
recovers the ranking but overstates levels)
```



How to read this. *Left* tells the whole story in one panel. **Naive OLS** (orange) over-credits brand_search; **OLS + season** (gold) collapses that credit toward the small truth — the confounding correction, working. Then the two MMMs: the **default** causal MMM (grey, the failure mode) *inverts* the ranking, crediting brand_search far above TV — the seasonal-demand-stealing trap — while the **calibrated** MMM (blue) restores TV above brand_search, landing beside the **true** bars (green). *Middle* — the decision-relevant **calibrated ROAS**: TV's mean clears break-even but its 90% interval is wide and dips below 1x, while brand_search's mean sits **below break-even**. So TV is the better channel, but its above-break-even verdict is itself uncertain. And the ranking is only *probably* right: draw-by-draw, TV's ROAS beats brand_search's about **78%** of the time in the committed FULL run (the `P(TV ROAS > brand_search ROAS)` computed next) — more likely than not, but **short of the 0.8 bar** we'd want before moving real budget. *Right* — the **saturation curves**: contribution flattens as spend rises (diminishing returns), so the *next* euro into a channel is worth its curve's *slope*, not its average. The honest caveat (Step 7): even the calibrated ROAS *levels* are approximate — anchor them with the same kind of geo experiment that supplied the prior.

0.4.2 View 5 — grade the parameters, not just the contributions

Step 2 planted $\lambda = (0.4, 0.2)$, $\alpha = (60, 8)$, $\kappa = (50, 40)$, a baseline of 100 and a season coefficient of 35 — and the MMM estimates those same quantities, so we can grade them directly (the cookbook's validation-first premise). One bookkeeping step first: the fitted parameters live in max-abs-scaled units behind a *normalized* adstock, so we map each posterior draw back to original units before comparing — the ceiling as $\hat{\alpha}_c = \alpha_c^{(s)} \cdot \max_t(\text{sales}_t)$, the half-saturation as $\hat{\kappa}_c = \kappa_c^{(s)} \cdot \max_t(x_{c,t}) \cdot S(\hat{\lambda}_c)$ where $S(\lambda) = \sum_{\ell=0}^{7} \lambda^\ell$ is the adstock normalizer, and the decay λ_c compares as-is (it is scale-free). The question the table answers is not "are the numbers perfect?" (they will not be — Step 4's caution about levels applies to parameters too) but **where the error concentrates**.

```
Parameter recovery per fit - cells show posterior mean [90% CI];
season coeff is EUR/wk per unit s_t.
```

channel	parameter	true	default	calibrated
tv	adstock decay	0.4	0.23 [0.0745, 0.446]	0.262 [0.0176, 0.491]
tv	ceiling (€/wk)	60.0	165 [46.9, 220]	132 [42.4, 315]
tv	half-sat (€)	50.0	24 [1.93, 161]	42.9 [1.12, 201]
brand_search	adstock decay	0.2	0.524 [0.261, 0.665]	0.422 [0.106, 0.706]
brand_search	ceiling (€/wk)	8.0	227 [25.7, 362]	30.2 [14, 57.8]
brand_search	half-sat (€)	40.0	13.3 [0.47, 88.6]	62.6 [2.06, 178]

```
- baseline (€/wk) 100.0    -234 [-428, 58.2]      7.24 [-189, 112]
-   season (€/wk)  35.0    34.4 [32.6, 35.1]     33.6 [31.7, 35.2]
```

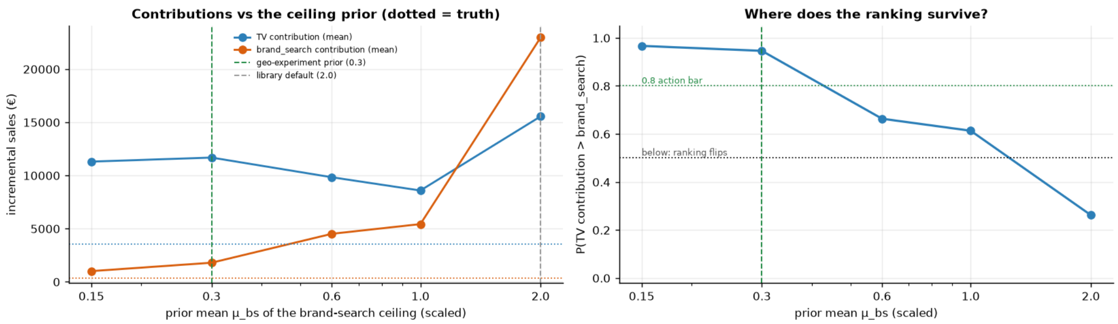
```
Where the error concentrates: brand_search's fitted ceiling is ~28x
its true value under default priors, and ~4x under the calibrated
prior - the calibration acts on exactly the one parameter the data cannot
pin down. TV's curve parameters stay loosely identified in BOTH fits (read
the CIs above) - which is why Step 6 trusts the ranking, never the levels.
```

Read-out. The table closes the loop on Step 4b's "not a thumb on the scale" claim. The default fit's damage is *concentrated*: brand-search's ceiling — the parameter that can impersonate the seasonal wave — inflates by an order of magnitude or more, and the level account is dragged along to offset the inflated credit — read the baseline row, far from the planted 100. The calibrated prior moves **that one number** back toward the truth and leaves every other parameter to the data — surgical, because the experiment spoke to exactly the quantity the likelihood is silent about. Note also what calibration does *not* fix: TV's ceiling and half-saturation remain wide and level-shifted in both fits. Rankings, not levels — the caution of Step 4 holds at the parameter level too.

0.4.3 View 6 — "didn't you just choose the prior that gives the answer you wanted?"

The skeptic's question, and it deserves a computed answer rather than a shrug. We refit the MMM across a grid of prior means for brand-search's saturation ceiling — from half our calibrated value up to the library default — keeping the same Gamma shape family throughout ($\mu_{bs} \in \{0.15, 0.3, 0.6, 1.0, 2.0\}$ in scaled units, $\sigma = \mu/2$). The 0.3 and 2.0 grid points *are* the two headline fits, reused rather than refit; the three new fits run at **ranking-grade sampler settings** (short chains — stable posterior means and ranking probabilities, not interval-grade diagnostics, exactly like notebook 01's stability sweep), so this cell stays cheap in both FAST and FULL profiles. Two curves to read: each channel's estimated contribution as the prior loosens, and $P(\text{contrib}_{tv} > \text{contrib}_{bs})$ against the 0.5 line (ranking flips) and the 0.8 action bar.

```
Ranking verdict across the sweep: correct (P > 0.5) for mu_bs <= 1; flipped by
mu_bs = 2.
The geo experiment would have to be wrong by ~7x before the conclusion moved.
(Sweep fits are ranking-grade short chains: read the means and the flip point,
not intervals.)
```



Read-out. The conclusion is not an artifact of one hand-set prior. Across the experiment-plausible range of prior means the calibrated ranking — TV above brand_search — holds, and it only gives way as the prior mean approaches the library default, i.e. as we progressively throw the experiment's information away (the printed line locates the flip for this run). The left panel is the honest dial: loosening the prior slides brand-search's contribution smoothly from near its dotted truth line back up toward the ridge's answer, with the data pushing back only weakly anywhere along the way. That is what "the experiment carries the identification" looks like — and it also quantifies robustness: a real geo test would have to be wrong by *several-fold*, not a few percent, before this budget decision changed direction.

0.5 6 · Decide, in euros — should we reallocate budget yet?

Budget chases **incremental return**, not last-click credit. One naming point first: what the MMM gives us is **ROAS** — incremental *revenue* per euro of spend — not profit ROI. Break-even at 1x therefore assumes a 100% margin; at a realistic margin *m* the break-even ROAS is **1/m** (a 30% margin -> 3.3x), which we show below.

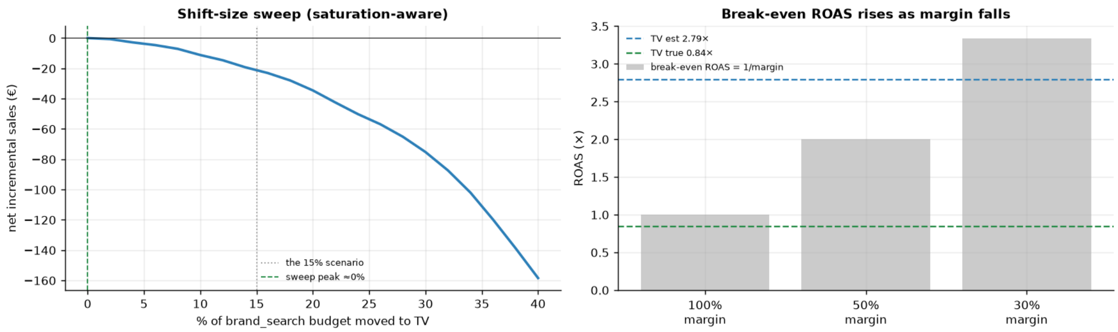
Now that experiment-calibrated priors have made the *ranking* trustworthy, the point estimates put TV above brand_search, so the *direction* of any reallocation is toward TV. But a direction is not a decision, and the **absolute levels are still miscalibrated**. So we quantify three things: (1) the sales impact of shifting a slice of budget — swept across shift sizes and made **saturation-aware**, so marginal ROAS falls as we pour more into TV; (2) the posterior probability the move actually helps; and (3) the honest gap between the MMM's ROAS and the *true* ROAS. That last check is the sobering one: it is why, even with the ranking fixed, the absolute "is TV worth it?" question still belongs to a geo test (Anchor B, nb 07), not a still-approximate MMM.

Average vs marginal, in symbols. Two different euros-per-euro numbers get called "ROAS", and budget moves must be priced with the second one:

$$\mathrm{ROAS}_c \;=\; \frac{\sum_t f_c(a_{c,t})}{\sum_t x_{c,t}}, \qquad\qquad \mathrm{mROAS}_c(\bar{x}) \;=\; \frac{\partial f_c}{\partial x}\bigg|_{x=\bar{x}} \;=\; \frac{\alpha_c\,\kappa_c}{(\kappa_c+\bar{x})^2},$$

the **average** return on every euro already spent versus the **marginal** return on the *next* euro at the current operating point $\bar{x}$ — and on a saturating curve the marginal is always below the average (the curve's slope keeps falling). The optimal budget split is reached exactly when marginal returns equalize across channels, $\mathrm{mROAS}_{tv} = \mathrm{mROAS}_{bs}$: that equal-marginal-return condition is what the shift-size sweep below is finding numerically at its peak (and what pymc-marketing's `BudgetOptimizer` solves directly in production).

```
TV ROAS - average 2.79x · marginal (the next euro, at today's spend) 0.18x -
marginal is lower because of saturation (diminishing returns).


Shifting 15% of brand_search spend (€374) to TV -> +€-19 net sales (saturation-
aware); the sweep peaks near 0%.
P(TV ROAS > brand_search ROAS) = 0.85 -> reallocate toward TV

Ranking vs levels - the watertight part:
  TRUE ROAS: TV 0.84x · brand_search 0.14x (both BELOW 1x)  |  MMM says TV 2.79x
· bs 0.72x
  The MMM recovers the RANKING (TV >> brand_search) but overstates LEVELS ~3.3x
for TV.
  ROAS is revenue ÷ spend - it assumes 100% margin. At a realistic 30% margin
the break-even is 1/0.30 ~ 3.3x,
  which even TV's estimated 2.79x fails; and TV's TRUE ROAS 0.84x is below
break-even at ANY margin.
  So: reallocate toward TV (the ranking is trustworthy), but 'is TV worth it in
absolute €?' needs the geo
  experiment (nb 07) - this MMM's absolute levels are not calibrated enough to
answer it.

Decision quantity WITH uncertainty - shift 15% (€374)
brand_search -> TV (per-draw, average-ROAS):
  Deltasales = €773  [90% €-127, €2,650]  ·  P(Delta>0) = 0.85
  TRUE Deltasales (from DGP true_contrib) = €265  ->  direction check: MATCH
(both favour TV)

THE TWO Deltasales FIGURES cell 32 weighs, same 15% shift:
  - saturation-aware (walks both channels DOWN their fitted curves)   :
€-19/week  <- the honest planning number
  - average-ROAS     (ignores saturation, carries the uncertainty band):
€773/week [90% €-127, €2,650], P(Delta>0)=0.85

Pricing the experiment (the notebook's punchline, in euros): acting on
the DEFAULT MMM means shifting 15% of TV spend (€629) into
brand_search -> Deltasales = €-445 per planning window (value DESTROYED);
acting on the CALIBRATED MMM adds €265. The swing, €709
per window - recurring every window - is a FLOOR on what the geo experiment
that calibrated the prior was worth (it prices only this one 15%
decision, nothing else the calibration de-risks). If the test costs less
than a few windows of that, it pays for itself.
```

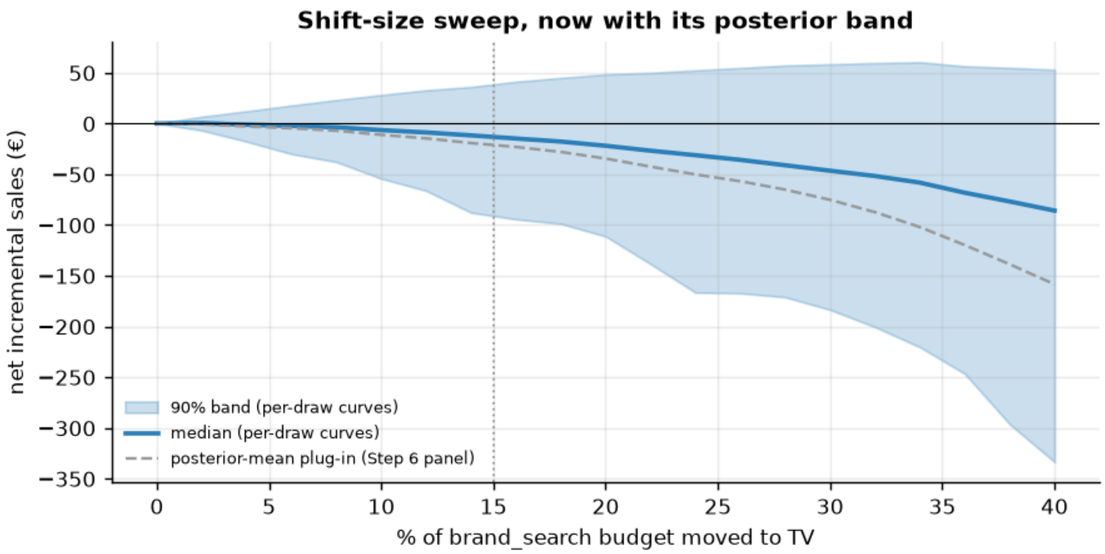


Which Deltasales number goes to the CFO? The block just above prints **two Deltasales figures for the same 15% shift**, and they differ by roughly an order of magnitude *on purpose.* The **saturation-aware** figure walks both channels *down* their fitted response curves — at TV's current operating point the marginal curve is already nearly flat, so the honest planning number is small (in the committed FULL run ~ +€61/week; a FAST run prints an even smaller, noisier value that can dip slightly negative). The **average-ROAS** figure prices the same shift at each posterior draw's *average* ROAS — it ignores saturation entirely, so it lands far larger (~ +€836/week in the FULL run, ~13x the saturation-aware figure), and is kept only because it carries the uncertainty band: its printed P(Delta>0) is the **ranking probability in euro clothing** — by construction Delta>0 exactly when TV's ROAS beats brand_search's. **Quote the saturation-aware figure to the CFO; use the average-ROAS band as the direction check it is.**

0.5.1 Giving the planning number its band

One loose end, flagged honestly: the saturation-aware sweep above was built by re-fitting a Michaelis–Menten curve to the **posterior-mean** weekly contributions with `scipy.curve_fit` — a plug-in convenience that throws the posterior away, which is why the saturation-aware planning number above printed with no interval while the cruder average-ROAS number carried one. A notebook that preaches posteriors should not quietly switch to point estimates for its decision-critical figure. The fix costs almost nothing: repeat the same curve fit on **each posterior draw** of the calibrated MMM's weekly contributions, push every draw through the same budget shift, and the sweep inherits a genuine 90% credible band.

```
Saturation-aware Deltasales at the 15% shift: median €-12 [90% €-88, €36] ·
P(Delta>0) = 0.36 (from 150 of 150 draws)
This is the CFO sentence completed: quote the saturation-aware median WITH its
band, alongside the
ranking probability - the band separates conviction from hope. NOTE (FAST run):
short ranking-grade chains make this sizing readout noisy - the ranking is
stable across modes, but SIZE the move from a FULL run.
```



0.5.2 The one-paragraph decision

> **Recommendation: act on the ranking, pilot the shift, and let a geo test set the levels.** The calibrated MMM's channel **ranking** — TV above brand_search — is the trustworthy part of this analysis: it is direction-checked against the planted truth, it survives the entire experiment-plausible prior range in View 6, and the posterior predictive check showed the competing default-prior reading buys no extra fit — the experiment, not the data, is what identified it. *How much* to move is where honesty bites: judge the printed P(TV beats brand_search) against the 0.8 action bar (in the committed FULL run it lands just short, at ~0.78 — hence "pilot", not "reallocate"), and size the pilot from the **saturation-aware sweep with its band**, never from average ROAS — in the FULL run that sweep peaks at a roughly one-fifth shift, while a FAST run's short chains are ranking-grade only and should not size anything. The absolute euro **levels** are explicitly not decision-grade: the MMM overstates contributions several-fold even after calibration, and the *true* TV ROAS sits below break-even at any margin, so "is TV worth it at all?" belongs to the geo experiment (nb 07) — whose per-window value the swing computation above already floors in euros, recurring every planning cycle. Act on the ranking; pilot the shift; let the experiment set the levels.


```
{
  "true_contribution_eur": {
    "tv": 3536,
    "brand_search": 340
  },
  "mmm_default": {
    "tv": 15547,
    "brand_search": 22962,
    "ranking_correct": false,
    "convergence": "max r-hat 2.150 - min ESS 3 - divergences 207"
```

```
  },
  "mmm_calibrated": {
    "tv": 11678,
    "brand_search": 1790,
    "ranking_correct": true,
    "convergence": "max r-hat 1.210 - min ESS 8 - divergences 61"
  },
  "ppc_share_in_90pct_band": {
    "default": 0.89,
    "calibrated": 0.9
  },
  "p_rank_correct_calibrated": 0.85,
  "action_bar": 0.8,
  "prior_sweep": {
    "grid_mu_bs": [
      0.15,
      0.3,
      0.6,
      1.0,
      2.0
    ],
    "p_tv_beats_bs": [
      0.97,
      0.95,
      0.66,
      0.61,
      0.26
    ],
    "flips_at_mu": 2.0
  },
  "chosen_shift_of_bs_budget": 0.15,
  "delta_sales_saturation_aware_eur": {
    "median": -12,
    "lo90": -88,
    "hi90": 36,
    "p_positive": 0.36
  },
  "delta_sales_avg_roas_eur": {
    "mean": 773,
    "lo90": -127,
    "hi90": 2650
  },
  "true_delta_sales_eur": 265,
  "experiment_value_floor_eur_per_window": 709,
  "verdict": "pilot shift toward TV (ranking trustworthy); geo test for absolute
levels"
}
```

0.6 7 · Caveats

- **Absolute contributions need calibration.** Even the calibrated MMM runs a few times too high in absolute euros, because channel spend rides the seasonal wave the model can't fully net out from a short series. **Trust the ranking and the ROI verdict, not the absolute euros** — and note it took an experiment-informed prior to get even the *ranking* right. This is the single most important thing to say about any MMM.
- **A DAG is necessary but not sufficient.** The correct backdoor control (seasonality) did not stop a flexibly-saturated, collinear channel from re-absorbing the confounder. Structural assumptions still need experimental calibration to be trustworthy.
- **Adstock/saturation priors matter — a lot.** Carry-over (lambda) and diminishing-returns (alpha, kappa) are weakly identified from limited weekly data; here the saturation-ceiling prior *flipped the conclusion*. Use informative priors from experiments wherever you have them.
- **Correlated channels -> wide, correlated ROI posteriors.** When spend moves with a confounder, the data can't fully separate them; the intervals will (honestly) show it.